

Library Management System - Project Report

1. Title

Library Management System

2. Abstract

This project is a **Java Swing-based Library Management System** designed to manage books, borrowers, and transactions in a library. It allows the user to add, view, borrow, return, and search books efficiently using a graphical user interface (GUI).

3. Objectives

- To develop a user-friendly system for managing library books.
 - To provide a simple interface for borrowing and returning books.
 - To allow searching of books by title or author.
 - To preload Computer Science books to simulate a real library environment.
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4. Tools and Technologies

- Java SE (JDK 8 or higher)
 - Java Swing (for GUI)
 - IDE: Eclipse / IntelliJ IDEA
 - Operating System: Windows / Linux / MacOS
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5. Features

1. **Add Book:** Enter details of new books and store them in the system.
 2. **View Books:** Display all books in a tabular form including availability status.
 3. **Borrow Book:** Borrow books for a borrower and update availability.
 4. **Return Book:** Return borrowed books and update availability.
 5. **Search Book:** Search books using title or author keywords.
 6. **About Page:** Information about the system.
 7. **Sidebar Navigation:** Switch between different pages seamlessly.
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6. Preloaded Computer Science Books

- Introduction to Algorithms – Cormen
 - Artificial Intelligence: A Modern Approach – Russell
 - Clean Code – Robert C. Martin
 - Computer Networks – Tanenbaum
 - Data Structures and Algorithms in Java – Goodrich
 - Operating System Concepts – Silberschatz
 - Database System Concepts – Korth
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7. Implementation

The project is implemented using **Java Swing** and organized using **CardLayout** to allow multiple functional pages in one window. The main pages include:

- **Home/Dashboard**
- **Add Book**
- **View Books**
- **Borrow Book**
- **Return Book**
- **Search Book**
- **About Page**

The system maintains lists of Book and Borrower objects to manage data and provides interactive GUI elements for user interaction.

8. Project Structure

LibraryManagementSystem/

```
|
|
|  └─ src/
|
|    └─ library/
|
|        └─ LibraryManagementSystemUI.java
|
|        └─ Book.java
|
|        └─ Borrower.java
|
```

```
└─ resources/
|   └─ image.png
|
└─ README.md
```

9. Screenshots

(Include screenshots of: Home Dashboard, View Books Page, Add Book Page, Borrow/Return Pages, Search Book Page, and About Page.)

10. Conclusion

The Library Management System demonstrates the following:

- Functional GUI for library operations.
- Multi-page navigation using **CardLayout**.
- Management of books and borrowers efficiently.
- Interactive borrowing, returning, and searching of books.

This system can be further enhanced by adding database integration for persistent storage.

11. Author

Akanksha Soni

12. References

- Java Swing Documentation
- Oracle Java Tutorials
- Various online resources for GUI design and CardLayout usage